

DEVELOPING E-LEARNING CONTENT FOR GOVERNMENT SCHOOLS

**A Community Engagement Project
Report**

Submitted in partial fulfillment of the
Requirements for the award of the Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

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Abstract

Government school classrooms frequently face constraints in delivering uniform, interactive digital educational content due to fragmented resources, varying teacher technical experience, and limited centralized tools. This Community Engagement Project (CEP) presents the development of a web-based e-learning platform specifically designed to empower teachers in government schools to structure, manage, and distribute digital educational content to students efficiently.

The platform provides a dual-role framework separating content creation responsibilities from student access. Teachers are equipped with intuitive authoring tools to create subject topics, organize sequential page-by-page lesson modules, embed explanatory text, upload educational videos (MP4 format), and attach supplementary reference materials in PDF format. Furthermore, an integrated quiz engine enables teachers to construct topic-specific assessments with multiple-choice questions to evaluate student comprehension.

To preserve academic integrity and prevent accidental modification of curriculum materials, the system enforces strict role-based access control. Students log into dedicated student portals where they can view assigned topics, study embedded notes, stream videos, read PDF documents, and attempt quizzes. Students are explicitly restricted from editing, deleting, or altering any lesson content or quiz structures.

By establishing a structured, centralized repository for multimedia learning materials, this project improves learning accessibility, supports teachers in organizing digital content, and provides a scalable tool for community-focused educational enhancement in government schools.

ACKNOWLEDGEMENT

I take this opportunity to express my profound gratitude and indebtedness to our project guide Harmanpreet Kaur for giving me the opportunity to accomplish this project. Her continuous guidance, insightful feedback, and constant encouragement throughout the development of this project were invaluable in overcoming technical and structural challenges.

I am very much thankful to our Principal Dr. Swati Vitkar for their kind co-operation and institutional support in the completion of my project.

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Finally, I wish to thank our lab assistant and all my friends and the entire Information Technology Department who directly helped me in the completion of this project. Last but not the least, I would like to express my sincere appreciation to my family without whose support, motivation and encouragement this effort would not have been possible.

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DECLARATION

I the undersigned Miss./Master PREM RAMKUMAR MANDAL, YASAR SHAIKH, SHIVAM RAI, SHUBHAM hereby, declare that the work embodied in this project work titled “Developing E-Learning Content for Government Schools” forms my own contribution to the research work carried out under the guidance of Harmanpreet Kaur is a result of my own research work and has not been previously submitted to any other University for any other Degree/ Diploma to this or any other University.

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Sr. No.	Date	Signature of Internal Guide
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1. INTRODUCTION

1.1 Purpose

The primary objective of this Community Engagement Project (CEP) is to design, develop, and document a structured, web-based educational content management system tailored for government schools. In many public educational institutions, teachers face significant operational challenges when attempting to introduce digital learning resources into traditional classroom settings. Learning materials—ranging from printed syllabus worksheets, reference PDF files, and explanatory video lectures—are often distributed across disjointed platforms, physical files, or unorganized digital storage.

This project addresses this problem by providing a unified web platform where teachers can systematically create subject topics, build sequential page-by-page lesson modules, attach multimedia files, and publish interactive self-assessment quizzes. By standardizing digital content delivery, the platform assists teachers in delivering structured lessons while giving students a centralized, clear learning interface.

Crucially, the system implements strict role-based authorization to safeguard curriculum structure. Teachers possess exclusive content creation, modification, upload, and assignment privileges. Students interact with the platform through read-only learning portals where they can view assigned lessons, study attached PDF documents, stream video tutorials, and attempt quizzes without any capability to alter, corrupt, or delete educational materials.

1.2 Background Information

Government schools play a vital role in providing accessible primary and secondary education to students across diverse socio-economic backgrounds. In recent years, educational authorities have increasingly recognized the necessity of incorporating digital learning tools to supplement traditional textbook-based teaching. However, the adoption of digital education in public schools faces several practical challenges:

1. **Resource Fragmentation:** Teachers often collect high-quality educational videos and digital notes, but lack a single system to group these resources under relevant syllabus chapters.
2. **Technical Usability Barriers:** Complex commercial Learning Management Systems (LMS) often feature overwhelming configurations that require extensive technical administration, making them impractical for daily use by school staff.
3. **Data Security and Access Control:** Unrestricted digital sharing platforms carry the risk of students accidentally editing, overwriting, or deleting shared study files.

4. Assessment Tracking: Traditional paper worksheets require substantial manual grading time, reducing the frequency of quick formative self-assessments.

To resolve these challenges, a lightweight, role-restricted e-learning platform was conceptualized. The system emphasizes clear content organization, ease of lesson page creation, direct PDF/video embedding, and automated quiz scoring.

1.3 Scope of the Report

This report documents the end-to-end development methodology, system architecture, database design, user interface specifications, testing protocols, and educational observations of the project. The scope encompasses:

- Requirement analysis covering teacher content authoring, multimedia handling, student consumption, and access security.
- System architecture including Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), Use Case models, and workflow specifications.
- Functional implementation of teacher management modules, page-by-page lesson generation, PDF document viewing, video stream integration, and multiple-choice quiz evaluation engines.
- Security design verifying strict role separation between content creators (teachers) and content consumers (students).
- Analysis of system performance, workflow efficiency, content integrity, and strategic recommendations for future institutional expansion.

2. LITERATURE REVIEW & SOFTWARE REQUIREMENTS

2.1 Existing Systems and Solutions

A survey of contemporary educational platforms reveals a broad spectrum of digital content management systems ranging from enterprise-grade Learning Management Systems (e.g., Moodle, Canvas) to basic file-sharing repositories (e.g., Google Drive, shared cloud folders).

Enterprise LMS platforms offer comprehensive analytics, grade books, and multi-tenant administrative controls. However, their operational complexity, steep learning curve, and substantial infrastructure overhead often pose significant implementation hurdles for government school environments where dedicated IT support staff may not be readily available.

On the other hand, generic file-sharing cloud storage solutions allow fast document uploads but lack educational structure. Files are presented in generic folder trees without pedagogical sequencing, lesson context, or integrated self-assessment quizzes. Furthermore, setting granular permissions to ensure students cannot alter folder structures often leads to administrative friction for non-technical teachers.

The system developed in this project bridges this gap by combining the simple content organization of an educational binder with the structured security of an academic portal.

2.2 Role of Technology in Community Engagement

Community Engagement Projects (CEP) aim to address real-world community challenges through technology-driven interventions. In the context of public school education, digital tools serve as equalizers by providing students with consistent, structured learning materials regardless of external constraints.

When digital learning platforms are tailored to local school needs, they empower teachers to curate high-quality educational content and build reusable digital learning libraries. This collective repository benefits multiple student batches over successive academic years, reinforcing community-level educational sustainability.

2.3 Review of Relevant Technologies

The implementation of lightweight, responsive educational platforms relies on established web technologies and standards:

- HTML5 and CSS3: Provide semantic layout structures, responsive viewports, and native support for embedding video (<video>) and document (<iframe/object>) players without requiring third-party plugins.
- JavaScript (ES6+): Enables interactive client-side logic, dynamic DOM page manipulation, real-time quiz scoring, and asynchronous server requests.
- Service Worker API: Enables Progressive Web Application (PWA) features such as static asset caching, offline page access, and fast loading over unreliable network links.
- Relational Database Storage (SQLite / PostgreSQL): Ensures ACID-compliant structured data storage for user accounts, topic metadata, lesson page sequences, quiz questions, and student progress records.

2.4 Software Requirements Specification (SRS)

The Software Requirements Specification (SRS) establishes the technical boundary, non-functional performance benchmarks, and functional capabilities of the Shiksha Setu platform:

1. Performance & Speed: Page render response times must remain below 1.5 seconds under normal local network operations.
2. Reliability & Availability: System data persistence must prevent loss of created lesson modules during abrupt browser closure or server restarts.
3. Usability & Accessibility: The user interface must enforce high-contrast text ratios, clear typography, and simple navigation designed for non-technical users.
4. Security & Role Authorization: Unauthorized modification of published curriculum content by student sessions must be 100% prevented by server-side authorization guards.

3. METHODOLOGY & SYSTEM ARCHITECTURE

3.1 Requirement Analysis & Role Security

To ensure the system addresses actual classroom needs, functional requirements were categorized into distinct user roles and technical operational domains.

3.1.1 Teacher Functional Requirements

1. Authentication: Secure login mechanism distinguishing teacher accounts from student accounts.
2. Topic Management: Ability to create, edit, rename, and organize subject topics (e.g., Science, Mathematics, English) by grade level.
3. Page-by-Page Lesson Builder: Capability to construct ordered lesson modules comprising sequential content pages.
4. Multimedia Uploads: Direct upload support for PDF reference files, reading guides, and MP4 video tutorials.
5. Quiz Generator: Interface to compose multiple-choice questions, define answer choices, specify correct keys, and assign point values.
6. Class Assignment: Ability to assign created topics and lesson modules to student groups.

3.1.2 Student Functional Requirements

1. Student Portal Access: Simple login using student credentials.
2. Dashboard Overview: Display of assigned topics, pending reading materials, and available quizzes.
3. Content Viewer: Integrated rendering of lesson text, inline PDF document viewer, and HTML5 video player.
4. Interactive Quiz Attempt: Interface to answer quiz questions and submit responses for immediate evaluation.
5. Result View: Feedback showing quiz score, total marks, and correct answer explanations.

3.1.3 Access Control & Security Requirements

1. Read-Only Student Restriction: Students must NOT be permitted to add, edit, modify, or delete topics, lesson pages, uploaded PDFs, videos, or quiz questions.
2. Route Protection: Server-side middleware must intercept and reject unauthorized write HTTP requests originating from student user sessions.

3.2 Data Collection Method

System requirements were established through workflow analysis of standard government school teaching routines, examination of curriculum syllabus frameworks, and observations of resource management practices:

- Workflow Observation: Analyzing how teachers organize physical lesson plans, textbook chapters, supplementary worksheets, and board notes.
- Syllabus Structure Mapping: Mapping subject units into hierarchical digital entities (Subject -> Topic -> Lesson Pages -> Resources & Quizzes).
- Usability Feedback Iterations: Evaluating non-technical user interaction patterns to streamline user interface navigation.

3.3 System Design & Modeling Diagrams

The system architecture follows a clean multi-tier model separating presentation, application routing, business logic, and database persistence.

3.3.1 Architectural Overview

Figure 1 illustrates the high-level system architecture connecting user web interfaces to the central application server and data stores.

Figure 1: Architectural Overview of E-Learning Platform

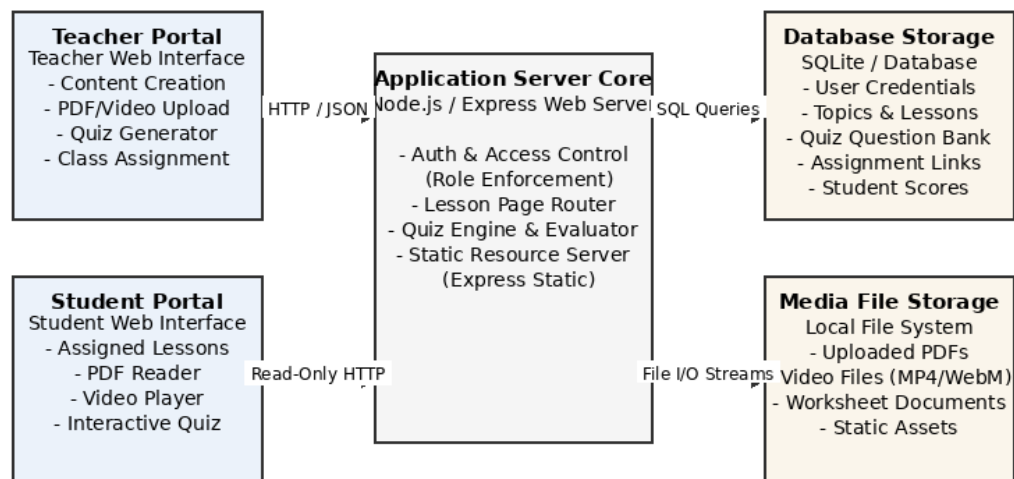


Figure 1: Architectural Overview of E-Learning Platform

3.3.2 Workflows

The operational processes for teachers and students are illustrated in Figures 2 and 3 below.

Figure 2: Teacher Content Creation and Assignment Workflow

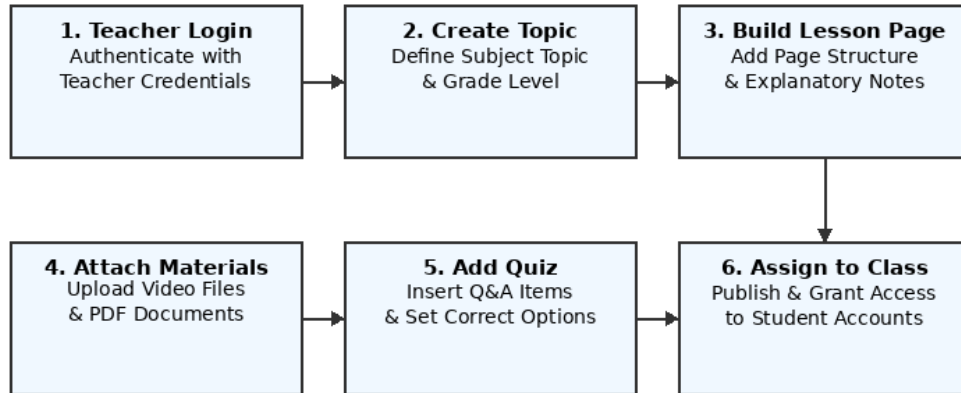


Figure 2: Teacher Content Creation and Assignment Workflow

Figure 3: Student Learning & Assessment Workflow

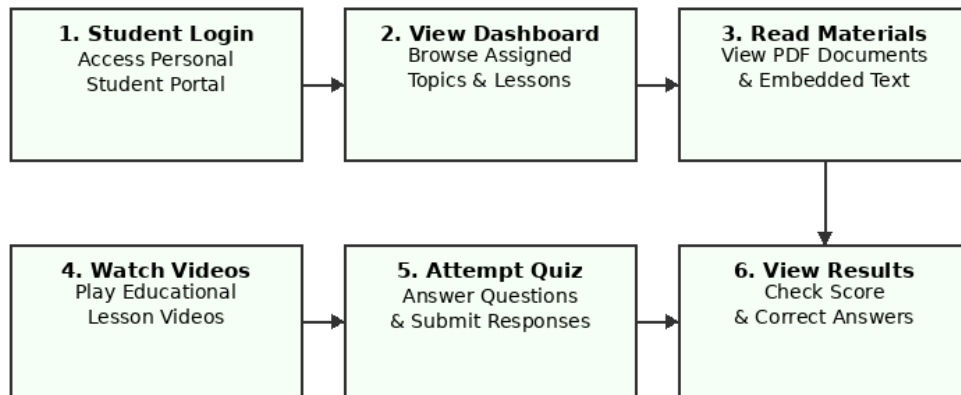


Figure 3: Student Learning & Assessment Workflow

3.3.3 Use Case Diagram

Figure 4 defines the system actor roles (Teacher vs. Student) and their respective interaction boundaries within the platform.

Figure 4: Use Case Diagram of E-Learning Content System

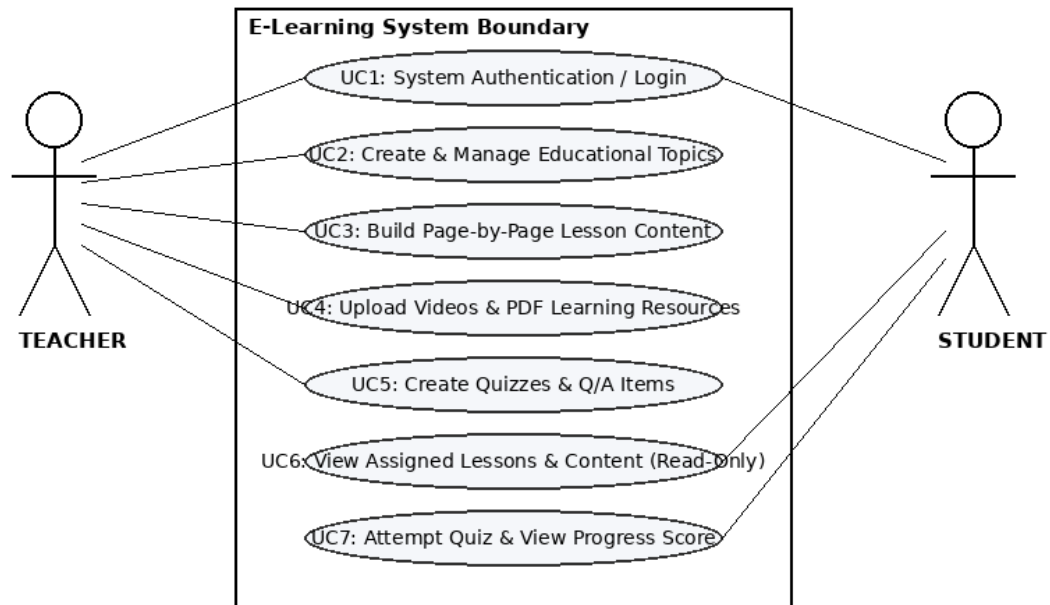


Figure 4: Use Case Diagram of E-Learning Content System

3.3.4 Data Flow Diagram (DFD)

Figure 5 details the Level 1 Data Flow Diagram tracing data movement from user inputs through application processes to persistent data stores.

Figure 5: Data Flow Diagram (DFD Level 1) - Content & Quiz Flow

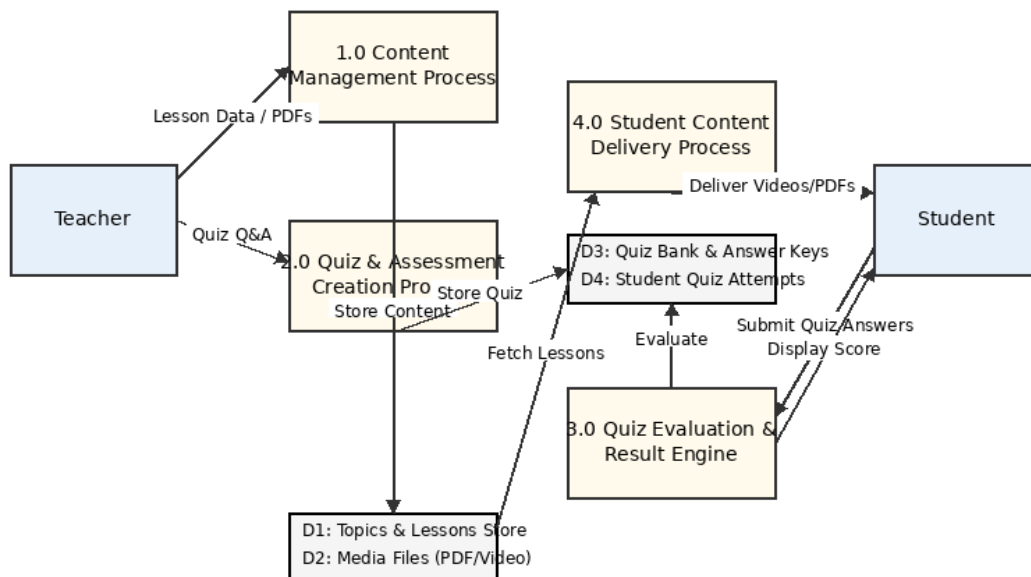


Figure 5: Data Flow Diagram (DFD Level 1) - Content & Quiz Flow

3.3.5 Entity Relationship Diagram (ERD)

The system database structure is defined in Figure 6, illustrating entities for User Accounts, Topics, Lesson Pages, Quizzes, and Questions.

Figure 6: Entity Relationship Diagram (ERD) of Database Schema

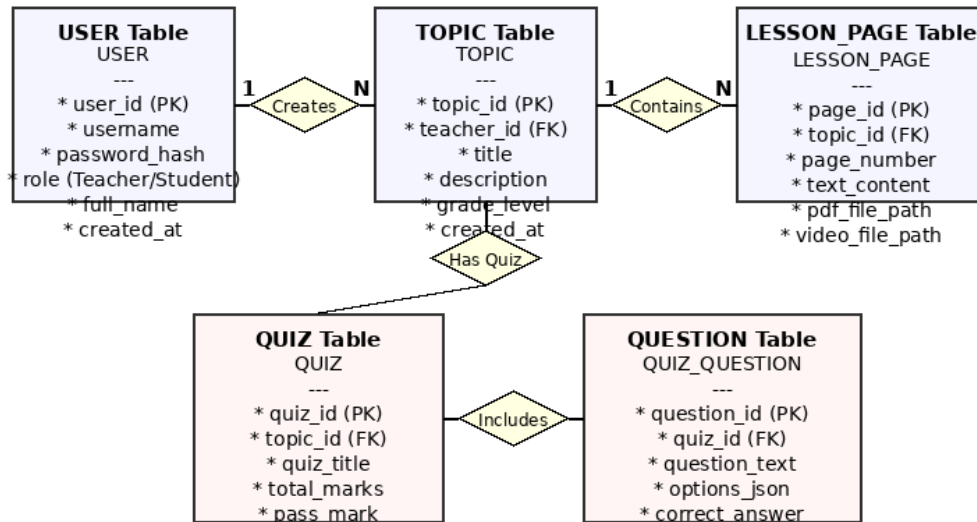


Figure 6: Entity Relationship Diagram (ERD) of Database Schema

3.3.6 Security Architecture & Access Control Flow

Figure 7 illustrates the server-side role enforcement mechanism guarding system routes against unauthorized content modification.

Figure 7: Access Control & Assignment Enforcement Architecture

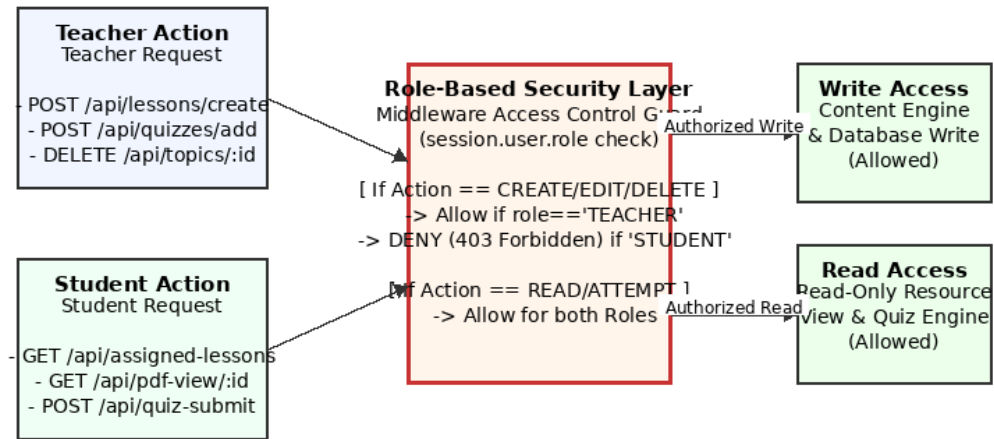


Figure 7: Access Control & Assignment Enforcement Architecture

3.4 User Interface Specifications & Screen Capture Walkthrough

The user interface of the Shiksha Setu platform has been designed for maximum clarity, clean visual hierarchy, and responsiveness. Figures 8 through 11 showcase the actual deployed website interfaces:

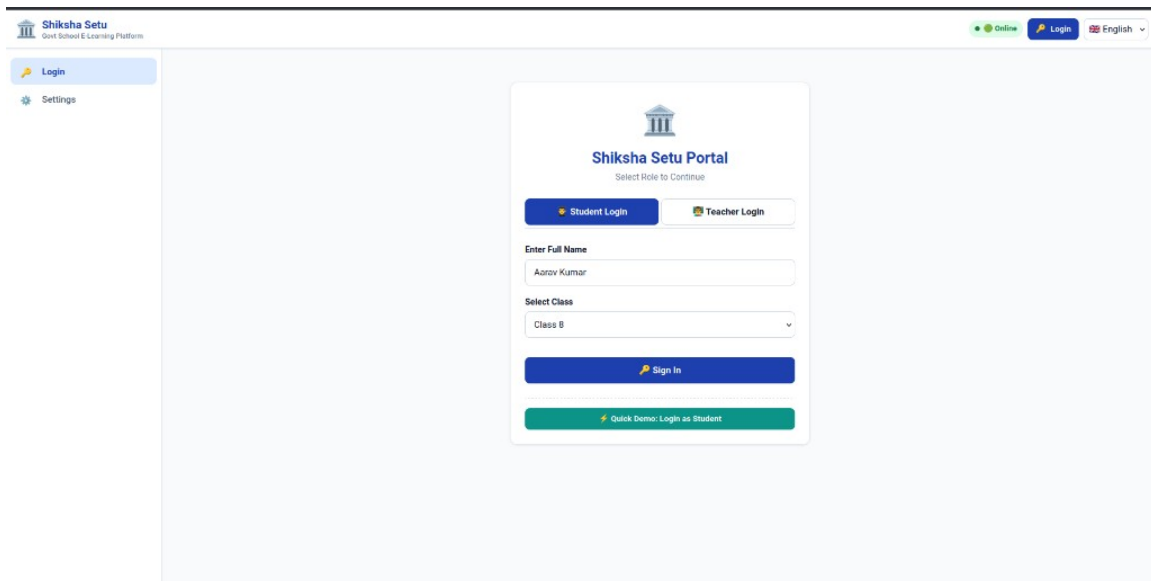


Figure 8: Shiksha Setu User Portal Login Screen (Student vs Teacher Role Switcher)

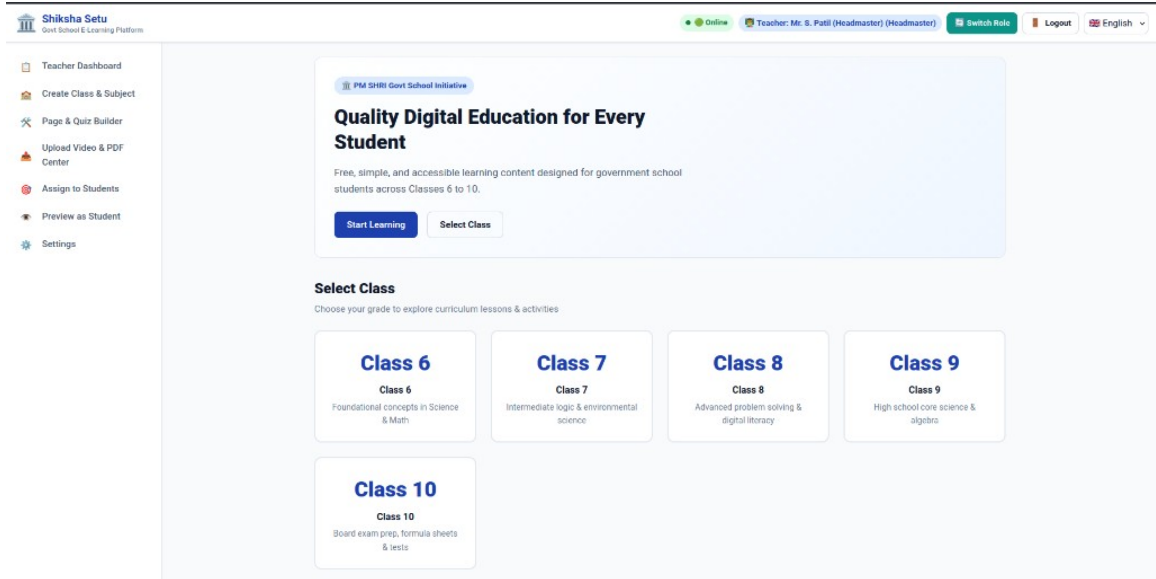


Figure 9: Student Dashboard & Class Selection Interface (Class 6 to 10)

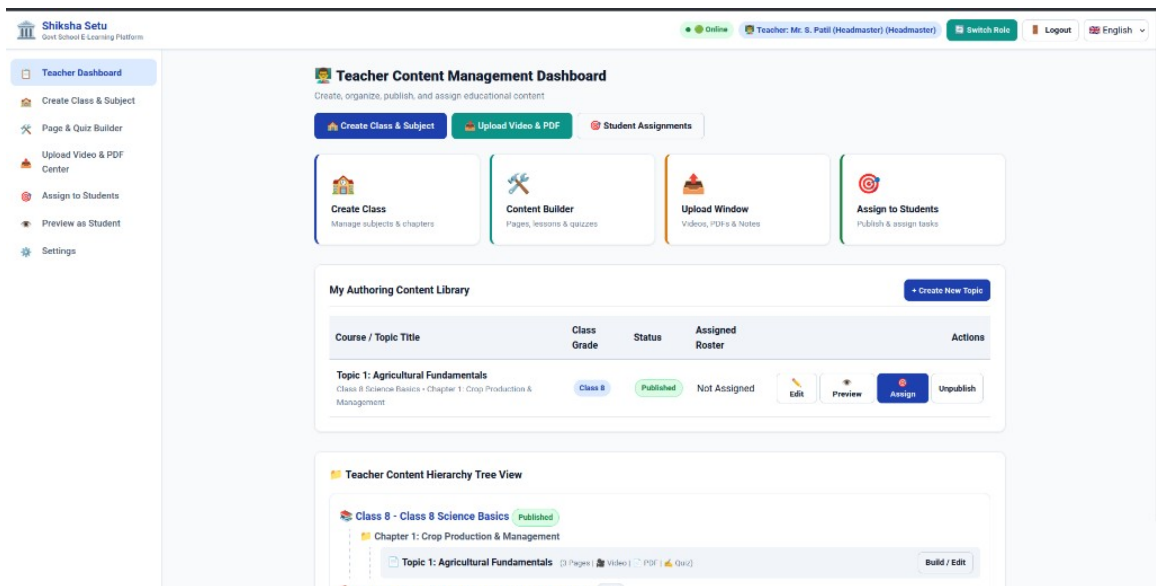


Figure 10: Teacher Content Management Dashboard & Hierarchy Tree View

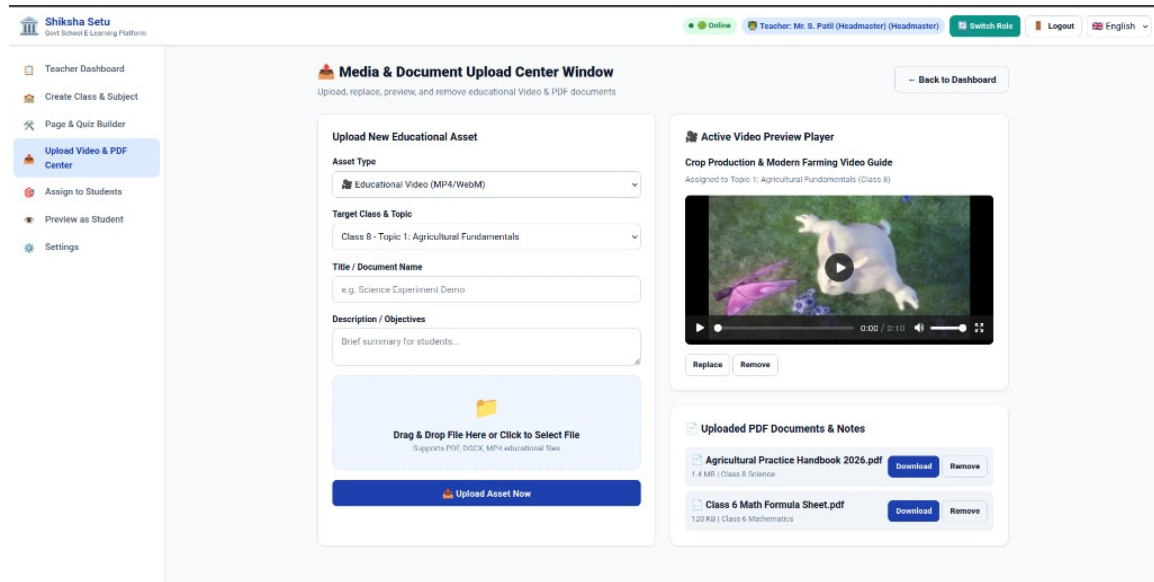


Figure 11: Media & Document Upload Center Window & Active Video Player

3.5 System Testing & Quality Assurance

Functional verification was conducted across all core modules to validate system behavior, media handling, and access security.

Table 3.1 details the functional test matrix executed during project evaluation:

Test ID	Feature Under Test	Test Procedure & Input	Expected Result
TC-01	Teacher Login	Submit valid teacher username & password	Successful login; redirect to Teacher Dashboard
TC-02	Topic Creation	Enter topic title 'Basic Physics' & grade level	New topic record created in database
TC-03	Lesson Page Add	Add Page 1 text notes & click save	Page created under selected topic
TC-04	PDF File Upload	Select & upload 'chapter1_notes.pdf'	File saved to media dir; link attached to page
TC-05	Video Upload	Select & upload 'gravity_demo.mp4'	Video uploaded; embedded player rendered
TC-06	Quiz Creation	Add 3 MCQ items with correct options	Quiz linked to topic; answer keys stored
TC-07	Student Login	Submit valid student credentials	Successful login; redirect to Student Portal
TC-08	Content View	Open assigned topic 'Basic Physics'	Lesson pages, PDF viewer & video render correctly
TC-09	Student Quiz Attempt	Select options for quiz & click submit	Responses evaluated; score calculated instantly
TC-10	Access Control Check	Student attempts POST /api/lessons/create	Server rejects request with 403 Forbidden

4. OBSERVATIONS AND ANALYSIS

4.1 Content Organization and Workflow Efficiency

The implementation of topic-wise, page-by-page lesson structuring demonstrated noticeable improvements in content organization compared to unorganized file repositories. Key observations include:

1. **Sequential Learning Sequence:** Grouping explanatory notes, PDF reference documents, and video clips on unified lesson pages provides students with a guided learning path, reducing confusion caused by searching through multiple independent files.
2. **Reduced Teacher Preparation Time:** Once a topic module is created, it can be assigned to multiple student batches without repeating upload steps, saving significant administrative effort.
3. **Centralized Media Storage:** Storing PDFs and videos within dedicated project storage ensures all learning materials remain accessible from any web browser.

4.2 Impact of Access Control on Educational Integrity

Enforcing strict role-based access control proved essential for maintaining curriculum integrity. During testing:

- Student portal sessions successfully prevented student accounts from accessing content editing tools, delete actions, or quiz configuration forms.
- Server-side middleware effectively blocked unauthorized HTTP requests, ensuring that even if a student attempted direct URL manipulation, administrative routes remained secured.
- This clear permission boundary provides teachers with complete confidence that curriculum materials cannot be altered or accidentally deleted by students.

4.3 Content Accessibility and Multimedia Delivery Analysis

Evaluating multimedia performance highlighted important operational insights:

- **PDF Document Rendering:** Native browser PDF rendering provided fast page load times and clear readability across desktop and tablet screen sizes.
- **Video Playback Performance:** HTML5 MP4 video playback offered smooth playback for educational clips. For optimal performance in low-bandwidth school environments, compressing video files prior to upload is recommended.

4.4 Quiz Engine Performance and Evaluation Accuracy

The automated quiz evaluation engine performed accurately during testing:

- Instant Grading: Student responses were evaluated immediately upon submission, providing students with prompt feedback on their understanding.
- Zero Grading Overhead: Automated scoring eliminates manual paper grading for teachers, allowing them to focus more time on classroom instruction.

5. CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The Community Engagement Project 'Developing E-Learning Content for Government Schools' successfully addresses the need for a structured, accessible, and secure digital educational platform tailored for public school environments. By delivering a web system that empowers teachers to create organized lesson pages, upload videos, attach PDF study guides, and generate interactive quizzes, the project enhances digital learning delivery.

Furthermore, the implementation of robust role-based access control ensures that educational integrity is preserved, giving students a focused, read-only portal to consume assigned content without compromising system security.

5.2 Limitations

While the system meets its core functional objectives, certain operational limitations exist:

1. Internet Connectivity Dependence: The platform requires an active network connection to access web pages and stream video media.
2. Media File Size Constraints: Uploading large high-definition video files can consume server storage and bandwidth rapidly.
3. Basic Quiz Formats: The current quiz engine primarily supports multiple-choice questions.

5.3 Recommendations

Based on system evaluation, the following implementation recommendations are suggested for schools adopting the platform:

- Video Optimization: Teachers should compress educational videos to standard resolution (480p/720p) before uploading to optimize bandwidth.
- Structured Topic Naming: Establish standard naming conventions for topics (e.g., 'Grade8_Science_Ch01') to maintain orderly archives.
- Regular Database Backups: Schedule periodic backups of the SQLite database and uploaded media directory.

5.4 Future Scope

Future enhancements to expand platform capabilities include:

1. Multilingual Support: Incorporating regional language interface toggles to support regional language government schools.

2. Offline Sync Mechanism: Developing progressive web app (PWA) offline caching for offline study on mobile devices.
3. Advanced Student Analytics: Adding analytical dashboards for teachers to track individual student lesson completion rates and quiz performance trends over time.
4. Low-Bandwidth Mode: Implementing automated media transcoding to adjust video streaming quality based on available network speeds.

ADDITIONAL DETAILED CONTENT

Shiksha Setu - Government School E-Learning Platform

The following pages expand the existing report using only the functionality and architecture supported by the submitted Shiksha Setu project files. No source-code listings and no application screenshots are included.

2.5 Detailed Software Requirements Specification

The Software Requirements Specification defines the expected behavior of Shiksha Setu from the perspective of students, teachers and the supporting technical system. The application is intended to make digital learning usable in conditions where device capability, network quality and language familiarity may vary. For that reason, the requirements are not limited to ordinary online content delivery. They also include local persistence, offline synchronization, accessibility controls and role-based security.

- Student accounts must support registration, login, profile access, class information, preferred language, learning progress, quiz history, XP and achievements.
- Teacher accounts must be separated from student accounts and must be authorized before they can create or modify classes, subjects, lessons, quizzes, assignments or library resources.
- The curriculum layer must organize learning material by class and subject so that students can navigate content appropriate to Classes 6 to 10.
- The system must support published lessons and quizzes, allowing students to consume learning content without receiving content-management permissions.
- The browser application must preserve important learner state locally so temporary network loss does not immediately interrupt the learning experience.
- Offline operations must be queued and synchronized when connectivity returns, while server-side records remain the persistent source for authenticated data.
- Interface text must support English, Hindi and Marathi, and accessibility preferences must remain persistent between sessions.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

2.6 Functional Requirements in Detail

The functional requirements describe the operations that a user can perform. They are derived from the project's route structure, frontend managers and relational database schema.

- Authentication: register and authenticate Student and Teacher users; return a signed token after successful authentication.
- Learning navigation: allow a learner to select a class, browse supported subjects, open lessons and move through educational content.
- Assessment: present chapter quizzes, maintain a timer, record selected answers, calculate results and save attempts.
- Progress: mark lessons complete, award XP, maintain streak information and expose completed lessons to the learner.
- Achievements: define badges and associate unlocked badges with individual students without creating duplicate ownership records.
- Teacher management: create and manage classes, subjects, lessons, quizzes and assignments through teacher-protected routes.
- Digital library: publish class- and subject-related study resources and allow students to retrieve published material.
- Synchronization: push queued offline operations and pull server state after connectivity is restored.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

2.7 Non-Functional Requirements in Detail

Non-functional requirements determine how well the system should operate rather than which individual buttons it provides. In Shiksha Setu these requirements are especially important because the intended learning context includes mobile devices, intermittent connectivity and users with different accessibility needs.

- Usability: navigation should remain simple and understandable for school learners and teachers who may have limited technical experience.
- Accessibility: the interface should provide visible keyboard focus, high-contrast presentation, adjustable font size and speech-based reading support.
- Performance: static application assets should be cacheable, and unnecessary visual effects can be disabled through low-data mode.
- Reliability: browser-side state and the offline queue should reduce the risk of losing learner actions during temporary disconnection.
- Security: protected endpoints must validate JWTs and enforce Student/Teacher roles and student ownership rules.
- Maintainability: frontend concerns and backend routes, controllers, middleware and database services should remain separated.
- Data integrity: relational constraints and unique keys should prevent duplicate progress, invalid class values and duplicate badge ownership.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

3.5 Detailed System Architecture

Shiksha Setu follows a layered web architecture. The presentation layer is implemented in HTML, CSS and JavaScript. The browser layer handles routing, language selection, accessibility settings, quiz interaction, local learner state and synchronization. The API layer is implemented with Node.js and Express and exposes versioned REST endpoints. The data layer uses PostgreSQL, with Supabase supported as a hosted PostgreSQL connection.

- Presentation layer: responsive pages, navigation, dashboards, lessons, quizzes, library views and teacher management views.
- Client application layer: application router, API client, progress tracker, quiz engine, language manager, accessibility manager and synchronization manager.
- Service/API layer: Express routes for authentication, students, teachers, progress, quizzes, badges, library resources and synchronization.
- Security layer: JWT verification, role authorization, ownership checking, request validation, CORS and rate limiting.
- Persistence layer: PostgreSQL tables for schools, users, curriculum, assessments, progress, badges, assignments and library resources.
- Offline layer: LocalStorage for persistent browser state, IndexedDB for pending operations and a service worker for cached application assets.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

3.6 Student Workflow

A Student begins by registering or logging in. After authentication, the learner can use the dashboard and curriculum navigation appropriate to the selected class. Lessons can be opened and completed, quizzes can be attempted, and the learner's progress can be reflected through XP, completed lessons, streak information and badges. When the browser is offline, supported learner actions are retained locally and synchronization can occur after the connection returns.

- Login or register as Student.
- Load the Student profile and selected class.
- Browse subjects and available lessons.
- Read lesson content and optionally use read-aloud accessibility support.
- Mark a lesson as completed and update progress.
- Attempt a timed quiz and submit answers.
- Review quiz score and achievement/progress information.
- Queue eligible changes while offline and synchronize them after reconnection.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

3.7 Teacher Workflow

The Teacher workflow is management-oriented. A valid Teacher token is required for protected teacher routes. Teachers can work with classes, subjects, lessons, quizzes, assignments and library resources. This separation is important because students should be able to consume learning material but must not be able to modify the published curriculum.

- Authenticate as Teacher.
- Open the teacher management dashboard.
- Create or manage classes and subjects.
- Create, update or publish lesson content.
- Create quizzes and associate them with learning content.
- Create assignments for classes.
- Add or manage digital library resources.
- Review student-oriented information available through authorized teacher endpoints.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

4.1 Student Module - Detailed Description

The Student module represents the primary learning experience. Student records include identifying and academic information such as class number and preferred language, together with learning-oriented values such as XP and streak. The frontend uses this state to provide a personalized dashboard and to present progress in a way that encourages continued participation.

- Class-aware curriculum access for Classes 6 through 10.
- Subject navigation across Mathematics, Science, English, Hindi, Social Science and Computer Basics.
- Lesson completion tracking using a student and lesson key relationship.
- Quiz history with score, percentage and time-taken information.
- Achievement display through badges associated with the student.
- Language and accessibility preferences stored for repeated use.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

4.2 Teacher Module - Detailed Description

The Teacher module provides the controlled authoring and management side of the platform. Teacher authorization is enforced on the backend rather than relying only on hidden frontend buttons. This is an important design choice because interface restrictions alone cannot protect server data.

- Teacher dashboard access through a protected route.
- Academic content creation for classes, subjects and lessons.
- Quiz creation and management.
- Assignment creation linked to classes and optionally lessons or quizzes.
- Digital library resource creation, update and deletion.
- Student roster or progress-oriented access where exposed by teacher-authorized endpoints.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

4.3 Quiz and Assessment Module

The assessment module converts learning content into measurable activity. The client quiz engine keeps track of the active quiz, current question, chosen answers and remaining time. On submission it calculates the number of correct responses and the percentage score. Authenticated attempts can be persisted through the backend so that quiz performance becomes part of the learner's recorded progress.

- A quiz may be linked to a lesson and represented by a stable quiz key.
- Questions can store multiple options, a correct option index and an explanation.
- The timer creates a bounded assessment session and can trigger automatic submission.
- Attempt records preserve score, total questions, percentage and time taken.
- Quiz completion can contribute XP and achievements such as the Quiz Master badge.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

4.4 Progress, XP, Streak and Badge Module

Gamification in Shiksha Setu is implemented through observable learning events. Completing lessons and attempting quizzes can contribute XP. Streak information represents repeated learning activity over time, while badges mark milestones. The database uses uniqueness rules so that the same student does not receive duplicate progress rows for one lesson or duplicate ownership of the same badge.

- Lesson completion is stored separately from the lesson content itself.
- XP earned for progress is constrained to non-negative values.
- Badges have a key, name, description, icon and requirement.
- Student badges store the time at which an achievement was unlocked.
- Example achievements include First Lesson, Quiz Master, 3-Day Streak, Math Genius and Science Explorer.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

4.5 Digital Library Module

The Digital Library provides reusable study resources outside the normal lesson sequence. A resource can be associated with a class and subject and can contain a title, description, type and URL. Publication status allows the system to distinguish material intended for learners from material that is not yet published.

- Resources can be filtered or organized by class and subject.
- Teacher ownership is represented through the uploaded-by relationship.
- Published resources can be retrieved by students.
- Teacher-protected operations can add, modify or remove library records.
- Seed data demonstrates notes, formula sheets, model question papers and study material.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

4.6 Multilingual Support

The language layer supports English, Hindi and Marathi. Translation keys are used for navigation, dashboard labels, quiz labels, library text, teacher-interface text, certificate text and settings. The selected language is retained locally, allowing the application to restore the learner's preference on a later visit.

- English uses the 'en' language code.
- Hindi uses the 'hi' language code.
- Marathi uses the 'mr' language code.
- Language changes are applied to interface elements using translation keys.
- A language-change event allows active views to re-render after the preference changes.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

4.7 Accessibility and Low-Data Design

Accessibility is integrated into the frontend rather than being treated as a separate page. The project includes font-size controls, high-contrast mode, low-data mode and text-to-speech. These features are particularly relevant to a learning portal because students may have different visual, reading or device constraints.

- Font-size choices modify the root document class and persist in LocalStorage.
- High-contrast mode changes design variables and strengthens visible borders.
- Low-data mode disables animations and transitions and can hide decorative imagery.
- Visible focus states support keyboard navigation.
- Speech synthesis can read lesson text using language-appropriate Indian locales.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

4.8 Offline-First and Synchronization Design

Offline capability is one of the most important technical features in the submitted implementation. The Sync Manager opens an IndexedDB database containing a pending-operations store. When the application is offline, eligible operations can be retained. When the browser reports that connectivity has returned and the user is authenticated, the manager can flush queued operations to the backend.

- IndexedDB database: shiksha_sync_db.
- Pending operation types include lesson completion, quiz attempt, XP addition, badge unlock and profile synchronization.
- The queue includes timestamps so operations can be processed in a meaningful order.
- An online event listener triggers synchronization after connectivity returns.
- An offline event listener updates the application state so the interface can reflect the connection change.
- If IndexedDB cannot be opened, the application degrades without preventing the LocalStorage-based experience.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.

5.1 Database Design Principles

The relational schema models the educational domain using linked tables rather than one large record. This reduces duplication and allows different types of information to evolve independently. Foreign keys preserve relationships between schools, users, curriculum entities and learner activity.

- Schools provide the institutional parent for teachers, students and classes.
- Classes organize school levels, while subjects and chapters organize curriculum structure.
- Lessons store learning units and publication status.
- Quizzes and quiz questions separate assessment metadata from individual questions.
- Quiz attempts and student progress store learner activity without changing curriculum records.
- Assignments link teacher activity to classes and optionally to lessons or quizzes.
- Library resources store supplementary study material.

Project relevance: This section is based on the submitted Shiksha Setu implementation and expands the explanation for academic-report purposes without reproducing source code or adding output screenshots.